Felix-Cosmin Mocanu

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fcmocanu

computational electrochemistry • molecular dynamics • phase-change memory materials • thermal transport • machine-learned interatomic potentials • quantum-dots • non-equilibrium and glassy dynamics

EDUCATION

2015–2018 **PhD Chemistry**, *Department of Chemistry*, University of Cambridge, *2018 Outstanding Thesis Prize Winner*

2014–2015 **MPhil Scientific Computing**, *Centre for Doctoral Training in Computational Methods for Materials Science*, University of Cambridge, *Distinction*

2010–2014 **MSci Natural Sciences**, *Department of Chemistry*, University of Cambridge, *Upper Second Class (Hons)*

RESEARCH

Postdoc **Computational modelling of Li-ion battery cathodes, (2023–)**

Postdoctoral Research Fellow in the Materials Modelling Laboratory (MML), Department of Materials at the University of Oxford. Working in the group of Prof. Saiful Islam on the computational modelling of next-generation Li-ion battery cathode materials. Member of the CATMAT collaboration of the Faraday Institution.

Postdoc **Low-temperature excitations in glasses, (2020–2023)**

Postdoctoral Research Fellow in the group on Disordered Systems and applications (Francesco Zamponi) at Laboratoire de Physique de l'École normale supérieure *(ENS, Paris)*

Working on numerical models of tunneling two-level systems in glasses as part of the ERC project "GlassUniversality" and as a member of the Simons Collaboration on "Cracking the Glass Problem".

Postdoc **Atomistic simulations of materials for quantum-dot LED and smart textile applications, (2018–2020)**

Research Assistant in the Nanoscience and Technology Group (Prof. Jong Min Kim) at the Centre for Advanced Photonics and Electronics *(University of Cambridge)*

Modelling semiconductor alloys for quantum dot light emitting diodes (QDLED). As part of our multidisciplinary team (ERC project 1D-Neon) I worked on the challenges of fiber-based electronic devices from materials to integrated smart textile systems.

PhD **First and second principles simulations of Ge-Sb-Te phase-change memory materials**

with Prof. Stephen R. Elliott and Prof. Gábor Csányi *(University of Cambridge)*

Monte-Carlo sampling techniques for the fitting of a transferable Ge-Sb-Te machine-learned interatomic potential. Structure searching and prediction of novel compositions for phase-change memory and thermoelectric materials. Non-equilibrium molecular dynamics and stochastic-boundary conditions methods for calculating thermal conductivity.

MPhil **Towards high-throughput simulation of Ge-Sb-Te phase change materials**

with Prof. Stephen R. Elliott and Prof. Gábor Csányi *(University of Cambridge)*

Worked on a machine-learned Gaussian Approximation Potential for the $Ge_2Sb_2Te_5$ phase-change memory material trained from *ab-initio* molecular dynamics trajectories and its application in large-scale and high-throughput simulations.

MSci ***Ab-inito* Molecular Dynamics of Phase Change Materials - NAND gate simulation**

with Dr. James Dixon and Prof. Stephen R. Elliott *(University of Cambridge)*

Worked on the *ab-initio* molecular-dynamics simulations of the atomistic effects of thermal pulse programming in the $Ge_2Sb_2Te_5$ phase-change memory material for applications in-memory logic and neuromorphic ("brain-like") computation.

UG **Synthesis and characterisation of functional curcubituril-azobenzene host-guest assemblies**

summer student with Dr. Jesús del Barrio Lasheras and Prof. Oren Scherman *(University of Cambridge)*

INTERSHIPS

2012 **Rotatives Engineering Intern**

Cambridge Materials Placements for Undergraduates in Summer (CaMPUS) from the University of Camb ridge Materials Science Department with Dr.-Ing. Hartmut Schlums *(Rolls-Royce Deutschland)* studying turbine blade lifetime in jet engines.

2018 **PhD Intern**

Center for Doctoral Training in Computational Methods for Materials Science Internship at Dassault Systémes, Cambridge, United Kingdom - with Dr. Felix Hanke, making workflows for estimating ionic transport of Li ions in commercial liquid electrolytes.

TEACHING

2016–2018 Organiser of fortnightly simulation tutorials *(Chemical Physics Group)*.

2017 Demonstrating Part IB Computational Practicals *(Department of Chemistry)*.

2016 Demonstrating Part IA Labs *(Department of Chemistry)*.

2016– Supervisor of Part II Theoretical Techniques course in Chemistry (Churchill College).

2015 Demonstrator IA C++ Computer Practicals (Engineering Laboratory).

2014–2016 Organiser of simulation sub-group meetings *(Chemical Physics Group)*.

SIMULATION AND COMPUTING

DFT	**VASP**, CP2K, CASTEP, GPAW	MD	**LAMMPS**, QUIP
Packages	**ASE**, quippy, pymatgen	MLIP	**MACE**, CHGNet, GAP
Languages	**Python**, Julia, Fortran, C++	Platforms	**Linux**

System administrator for the Elliott group *Hathor* computer cluster.

Tier-2 Resource Allocation (Project T2-CS020 “Thermoelectric Properties of Disordered Materials”): awarded 4 million CPU hours on the CSD3 supercomputer with Yuchen Hu and Prof. Stephen R. Elliott.

Tier-2 Resource Allocation (Project T2-CS073 “First-principles simulations of nanocrystal early-stage growth and morphology”): awarded 3.5 million CPU hours and 25,000 GPU hours on the CSD3 supercomputer with Prof. Jong Min Kim.

MesoPSL resource allocation: awarded 500,000 CPU hours on the MesoPSL supercomputer with Dr. Francesco Zamponi at ENS.

Grand-challenge proposal “Understanding ion migration and stacking transitions in Li-rich layered cathode materials for high-energy-density batteries”: awarded 14 million CPU hours on the Archer2 UK National Supercomputer

PUBLICATIONS

Status: submitted, preprint, published.

23. Ricci, B., **Mocanu, Felix-Cosmin**, Chatzogiannakis, D., Fuentes, V., P. Black, A., Cluzeau, B., Tessier, C., Bernard, P., Saiful Islam, M. & Rosa Palacín, M. Investigating the asymmetric electrochemical, structural and electronic properties of Mn-rich Li(Mn,Fe)PO_4 electrode materials. *Journal of Materials Chemistry A* (2026).

22. Casella, J., Morzy, J., **Mocanu, Felix C.**, Müller, A., Yarema, M., Futscher, M. H., Islam, M. S. & Romanyuk, Y. E. Performance Comparison of Transition Metal (Cr, Mn, Fe, Co, Ni, Cu)-Fluoride Conversion Cathodes in Thin-Film Solid-State Batteries. *ACS Applied Energy Materials* **8,** 15101–15111. ISSN: 2574-0962, 2574-0962 (Oct. 2025).

21. Arber, A. N., Vikram, **Mocanu, Felix C.** & Islam, M. S. Ion Migration and Dopant Effects in the Gamma-$CsPbI_3$ Perovskite Photovoltaic Material: Atomistic Insights through *Ab Initio* and Machine Learning Methods. *Chemistry of Materials* (June 2025).

20. **Mocanu, Felix C.**, Elliott, S. R. & Konstantinou, K. Partial Melting and Structural Disorder in Models of Irradiated Amorphous $Ge_2Sb_2Te_5$. *physica status solidi (RRL) – Rapid Research Letters,* 2500037 (Apr. 2025).

19. Schirmacher, W., Paoluzzi, M., **Mocanu, Felix C.**, Khomenko, D., Szamel, G., Zamponi, F. & Ruocco, G. The nature of non-phononic excitations in disordered systems. *Nature Communications* **15,** 3107 (Apr. 2024).

18. Ciarella, S., Khomenko, D., Berthier, L., **Mocanu, Felix C.**, Reichman, D. R., Scalliet, C. & Zamponi, F. Finding defects in glasses through machine learning. *Nature Communications* **14,** 4229 (July 2023).

17. Hou, B., **Mocanu, Felix C.**, Cho, Y., Lim, J., Feng, J., Zhang, J., Hong, J., Pak, S., Park, J. B., Lee, Y.-W., Lee, J., Kim, B.-S., Morris, S. M., Sohn, J. I., Cha, S. & Kim, J. M. Evolution of Local Structural Motifs in Colloidal Quantum Dot Semiconductor Nanocrystals Leading to Nanofaceting. *Nano Letters* **23,** 2277–2286 (Mar. 2023).

16. **Mocanu, Felix C.**, Berthier, L., Ciarella, S., Khomenko, D., Reichman, D. R., Scalliet, C. & Zamponi, F. Microscopic observation of two-level systems in a metallic glass model. *The Journal of Chemical Physics* **158,** 014501 (Jan. 2023).

15. Konstantinou, K., **Mocanu, Felix C.** & Akola, J. Electron localization in recrystallized models of the $Ge_2Sb_2Te_5$ phase-change memory material. *Physical Review B* **106,** 184103 (Nov. 2022).

14. Konstantinou, K., **Mocanu, Felix C.**, Akola, J. & Elliott, S. R. Electric-field-induced annihilation of localized gap defect states in amorphous phase-change memory materials. *Acta Materialia* **223,** 117465 (Jan. 2022).

13. Konstantinou, K., Mavračić, J., **Mocanu, Felix C.** & Elliott, S. R. Simulation of Phase-Change-Memory and Thermoelectric Materials using Machine-Learned Interatomic Potentials: Sb_2Te_3. *physica status solidi (b)* **258,** 2000416 (Sept. 2021).

12. **Mocanu, Felix C.**, Konstantinou, K., Mavračić, J. & Elliott, S. R. On the Chemical Bonding of Amorphous Sb_2Te_3. *physica status solidi (RRL) - Rapid Research Letters* **15,** 2000485 (Mar. 2021).

11. Bang, S. Y., **Mocanu, Felix C.**, Lee, T. H., Yang, J., Zhan, S., Jung, S.-M., Shin, D.-W., Suh, Y.-H., Fan, X.-B., Lee, S., Choi, H. W., Occhipinti, L. G., Han, S. D. & Kim, J. M. Robust In−Zn−O Thin-Film Transistors with a Bilayer Heterostructure Design and a Low-Temperature Fabrication Process Using Vacuum and Solution Deposited Layers. *ACS Omega* **5,** 21593–21601 (Sept. 2020).

10. **Mocanu, F. C.**, Konstantinou, K. & Elliott, S. R. Quench-rate and size-dependent behaviour in glassy $Ge_2Sb_2Te_5$ models simulated with a machine-learned Gaussian approximation potential. en. *Journal of Physics D: Applied Physics* **53,** 244002 (Apr. 2020).

9. Krynski, M., **Mocanu, Felix C.** & Elliott, S. R. Elucidation of the Nature of Structural Relaxation in Glassy D-Sorbitol. *The Journal of Physical Chemistry B* (Feb. 2020).

8. **Mocanu, Felix C.**, Konstantinou, K. & Elliott, S. R. Nonequilibrium ab initio molecular-dynamics simulations of lattice thermal conductivity in irradiated glassy $Ge_2Sb_2Te_5$. *Applied Physics Letters* **116,** 031902 (Jan. 21, 2020).

7. Jung, S., Choi, H. W., **Mocanu, Felix C.**, Shin, D.-W., Chowdhury, M. F., Han, S. D., Suh, Y.-H., Cho, Y., Lee, H., Fan, X., Bang, S. Y., Zhan, S., Yang, J., Hou, B., Chun, Y. T., Lee, S., Occhipinti, L. G. & Kim, J. M. Modeling Electrical Percolation to optimize the Electromechanical Properties of CNT/Polymer Composites in Highly Stretchable Fiber Strain Sensors. *Scientific Reports* **9,** 20376 (Dec. 2019).

6. Hanke, F., Modrow, N., Akkermans, R., Korotkin, I., **Mocanu, Felix C.**, Neufeld, V. & Veit, M. Multiscale Electrolyte Transport Simulations for Lithium Ion Batteries. *Journal of The Electrochemical Society* **167** (May 2019).

5. Konstantinou, K., **Mocanu, Felix C.**, Lee, T.-H. & Elliott, S. R. Revealing the intrinsic nature of the mid-gap defects in amorphous $Ge_2Sb_2Te_5$. *Nature Communications* **10,** 1–10 (July 11, 2019).

4. **Mocanu, Felix C.**, Konstantinou, K., Lee, T. H., Bernstein, N., Deringer, V. L., Csányi, G. & Elliott, S. R. Modeling the Phase-Change Memory Material, $Ge_2Sb_2Te_5$, with a Machine-Learned Interatomic Potential. *The Journal of Physical Chemistry B* **122,** 8998–9006 (Sept. 27, 2018).

3. Konstantinou, K., **Mocanu, Felix C.**, Lee, T. H. & Elliott, S. R. *Ab initio* computer simulations of non-equilibrium radiation-induced cascades in amorphous $Ge_2Sb_2Te_5$. *Journal of Physics: Condensed Matter* **30,** 455401 (Nov. 14, 2018).

2. Mavračić, J., **Mocanu, Felix C.**, Deringer, V. L., Csányi, G. & Elliott, S. R. Similarity Between Amorphous and Crystalline Phases: The Case of TiO_2. *The Journal of Physical Chemistry Letters* **9,** 2985–2990 (June 7, 2018).

1. Konstantinou, K., Lee, T. H., **Mocanu, Felix C.** & Elliott, S. R. Origin of radiation tolerance in amorphous $Ge_2Sb_2Te_5$ phase-change random-access memory material. *Proceedings of the National Academy of Sciences* **115,** 5353–5358 (May 22, 2018).

CONFERENCE PRESENTATIONS

Faraday Institution Conference

2024 *Voltage fade in $Na_x[Li_{0.2}Mn_{0.8}]O_2$ an oxygen-redox cathode material*, Poster Presentation, University of Newcastle, United Kingdom

XVI International Workshop on Complex Systems

2023 *On the microscopic nature of two-level systems in glasses*, Contributed Talk, Andalo, Italy - Prize for the best talk by an early-career researcher

Physics of non-crystalline solids

2022 *A heart of glass - low-energy excitations in a fragile glass model*, Contributed Talk, University of Kent, United Kingdom

International Symposium on Doped Amorphous Chalcogenides and Devices

2018 *Computer simulations of phase-change materials*, Poster Presentation, Grasmere, United Kingdom - Runner-up Poster Prize

European Phase-Change and Ovonics Symposium

2017 *A machine-learned interatomic potential for Ge−Sb−Te phase-change materials*, Poster Presentation, RWTH Aachen, Germany

Psi-k Conference on Electronic Structure

2015 *A Gaussian Approximation Potential (GAP) for the $Ge_2Sb_2Te_5$ phase change material*, Poster Presentation, San Sebastián, Spain

REFEREES

Prof. Saiful Islam, University of Oxford; saiful.islam@materials.ox.ac.uk

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Dr. Bo Hou, University of Cardiff; HouB6@cardiff.ac.uk

Prof. Stephen R. Elliott, University of Cambridge; sre1@cam.ac.uk

Prof Gábor Csányi, University of Cambridge; gc121@cam.ac.uk